

Notification Delivery Systems

Research & Integration Methods — StockWise Project

What is a Notification Delivery System?

A notification delivery system alerts users about important events in real-time or on a schedule. In StockWise, notifications inform admins and staff when stock levels drop below a critical threshold.

Delivery Methods Compared

Method	Speed	Complexity	Use in StockWise
In-App Alert	Instant	Low	Dashboard banner / bell icon
Email (SMTP)	~5–30 sec	Low	Daily stock summary reports
SMS (Twilio)	Instant	Medium	Critical stockout alerts
WebSocket	Instant	High	Live dashboard push updates

Chosen Approach for StockWise

- In-App Notifications — primary method using the existing ALERTS table in PostgreSQL.
- Email via JavaMailSender (Spring Boot) — for daily/weekly stock summary reports.

How Integration Works

- Backend (Spring Boot) creates an Alert record when product quantity < threshold.
- Frontend (React) polls the /api/alerts endpoint every 30 seconds via Axios.
- Alert badge updates dynamically on the dashboard without page reload.

Security Notes

- SMTP/API credentials stored as environment variables — never in source code.
- Alert logs tracked with JPA Auditing (@CreatedDate) for full audit trail.